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9.

CO-2 AT-3 Class Test

1. Grey-Level Transformation

Gamma transformation is used to brighten dark regions of an image.

$$S = cr^{\gamma}$$

r is the normalized input intensity.

Steps :

*> Read the dark image and normalize pixel value.
*> $\gamma < 1$ such as 0.5
*> Apply the transformation to every pixel.

Result : Dark regions become brighter.

2. Histogram equalization

It removes image contrast by spreading intensity value accross a wide range.

Steps :

*> Calculate the histogram of the fingerprint image
*> Find the CDF.
*> Map each intensity using the CDF.

$$S_k = (L-1) \sum_{i=0}^{k} p(r_i)$$

3. Salt and Pepper Noise Removal

A median filter is suitable for removing salt and pepper.

Steps :

*> Select 3×3 neighbourhood around each pixel.

*> Arrange the nine pixel values in ascending order.

*> Repeat for all pixels.

Result : Noise is reduced.

4. Image Sharpening

Unsharp masking enhances edge and find details in an image.

Steps :

*> Blur the original image using a Gaussian filter.

*> Subtract the blurred image from original.

$$g(x,y) = f(x,y) + k\,[f(x,y) - f_D(x,y)]$$

Result : Copper tracks and their edges become sharper.

5. Frequency-domain Low-pass Filtering.

A Gaussian low-pass filter removes high-frequency noise and smooths the image.

$$H(u,v) = e^{-D^2(u,v)/(2D_0)^2}$$

Steps :

*> Apply the FT to image.

*> Multiply the frequency spectrum

*> Apply the inverse FT.

Result : Noise is reduced from become smoother.

6. High-pass Filtering

It enhances fine details and vessal edges.

Steps:

*> It convert the retinal image to the frequency.
*> Apply a high pass filter to emphasize high frequency.
*> Apply the inverse FT.

$$g = AF - f_D \qquad A > 1$$

Result: The bold vessel, and their edges become more visible.

7. Point detection and Line detection:

It isolated pixel, while line detection detects thin lines.

Point detection mask $\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$.

Horizontal $\begin{bmatrix} -1 & -1 & -1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{bmatrix}$

Steps:

*> Apply point mask to detect isolated air bubbles.
*> Mask pixels whose response exceeds the threshold.

Result: Bubbles and crack are identified.

8. Thresholding for Handwritten Answer sheet.

Adaptive threshold is suitable because of the illumination.

Steps:

*> Convert the image to grayscale.
*> divide the image into small local.

$$g(x,y) = \begin{cases} 1, & F(x,y) < T(x,y) \\ 0, & \text{otherwise} \end{cases}$$

Result: Hand written text is seperated from the background even under uneven light.

9. Region Growing for lake extraction

It give pixel with similar property.

Steps :

*> Select a speed point inside the lake.

*> Examine neighbouring pixel.

$$|T(x,y) - \mu_a| < T$$

10. Edge linking and Boundary Detection.

It connects broken edge segment with boundary.

*> Apply an edge detector such as canny detector.

*> Identify nearly edge pixel with similar gradient direction and magnitude.

*> Trace the connected contour and remove noisy segments.

Result : Discontinuous edge and linked to produce a more complete gear.